The empirical recurrence for OEIS A237483: recovering a conjecture that is not written down

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Abstract

OEIS A237483 records "Empirical recurrence of order 88", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 200$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 88, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 88 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A237483 is "Number of (n+1)X(2+1) 0..3 arrays with the maximum plus the minimum minus the lower median of every 2X2 subblock equal.". It has offset 1 and begins

$$1464, 15898, 180730, 2218016, 27849668, 361041844, 4724446414, 62643477220, \ldots$$

The entry states:

> Empirical recurrence of order 88 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 10:12:47, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 200$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 200.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 400$ exact terms of the model returns order 88 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{88} c_i \, a(n-i),$$

c_1	45	c_{23}	−550439090945393674478	c_{45}	−699402889001484792604403936663
c_2	−466	c_{24}	−1580714750717567899471	c_{46}	−328850683690156848152171131124
c_3	−8291	c_{25}	9997398366652665100912	c_{47}	2022514687539142781689785411270
c_4	199874	c_{26}	8862509715853352773442	c_{48}	−82691610413510556248930089566
c_5	−187243	c_{27}	−132618426190837951661077	c_{49}	−4624878728755303463792278403372
c_6	−27395034	c_{28}	45573433732151485025288	c_{50}	2067981221434891191342779845666
c_7	182275632	c_{29}	1332000090021718123599322	c_{51}	8358174498723274091763810015458
c_8	1759242718	c_{30}	−1761028637129243539787037	c_{52}	−6948011520668375504460115157148
c_9	−22897490225	c_{31}	−10155036510428624561906311	c_{53}	−1171751337154184436292646057908
c_{10}	−35479516029	c_{32}	23618263282616212702741106	c_{54}	15018195320790184236950942204932
c_{11}	1541240189465	c_{33}	56816699229242350837475982	c_{55}	12072251820061825798840257185376
c_{12}	−2983851254422	c_{34}	−212855924182543629774547151	c_{56}	−2410063841095473575337633363455
c_{13}	−64981162714459	c_{35}	−203418248756214265461846677	c_{57}	−7589635985905127931027839397600
c_{14}	299134055797142	c_{36}	1439190777130387271395538836	c_{58}	29887232040010272033485677455904
c_{15}	1725788095704285	c_{37}	109313557860524540794437314	c_{59}	−498361237357788350884753351136
c_{16}	−14047527762184836	c_{38}	−7562880679486620305507018671	c_{60}	−2889265797622467298608287054092
c_{17}	−23093251970849371	c_{39}	4428289812818603856867808151	c_{61}	8065455226749960423558071654656
c_{18}	432174961168292973	c_{40}	31135721431513151162164091141	c_{62}	21568739739719419077546289515328
c_{19}	−216036952443781705	c_{41}	−37124355274642720491006719912	c_{63}	−1130384804867235820080775480128
c_{20}	−9422204230197533218	c_{42}	−99096737186441885020214391834	c_{64}	−1205821700184426904177534422617
c_{21}	19672764107898696412	c_{43}	187887556215958642303441695092	c_{65}	9758806874418423987587524906496
c_{22}	147517260692207956513	c_{44}	231716994144473961250339295917	c_{66}	4671537151725735886904480174592

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 88, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $88 + 4$ terms removed returns order 88 as well, so $\deg q_{\min} = 88 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $88 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 88 that this sequence satisfies — and that family turns out to have exactly one member.

References

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